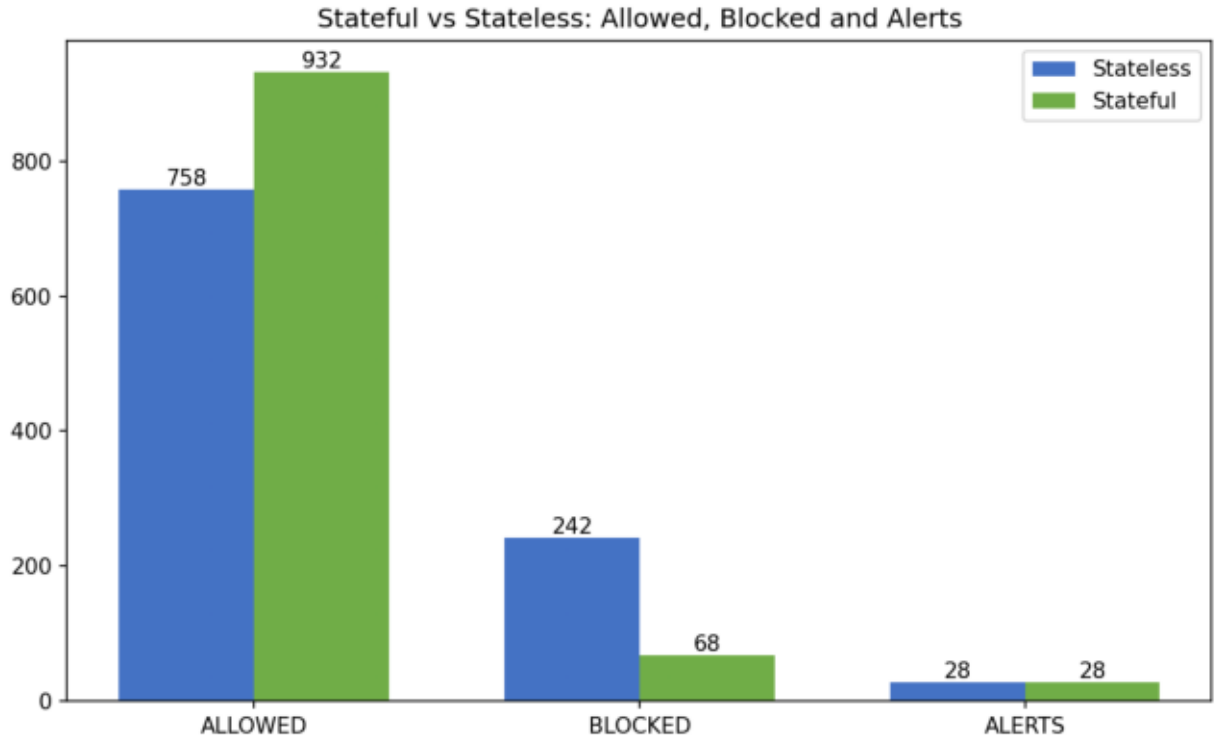
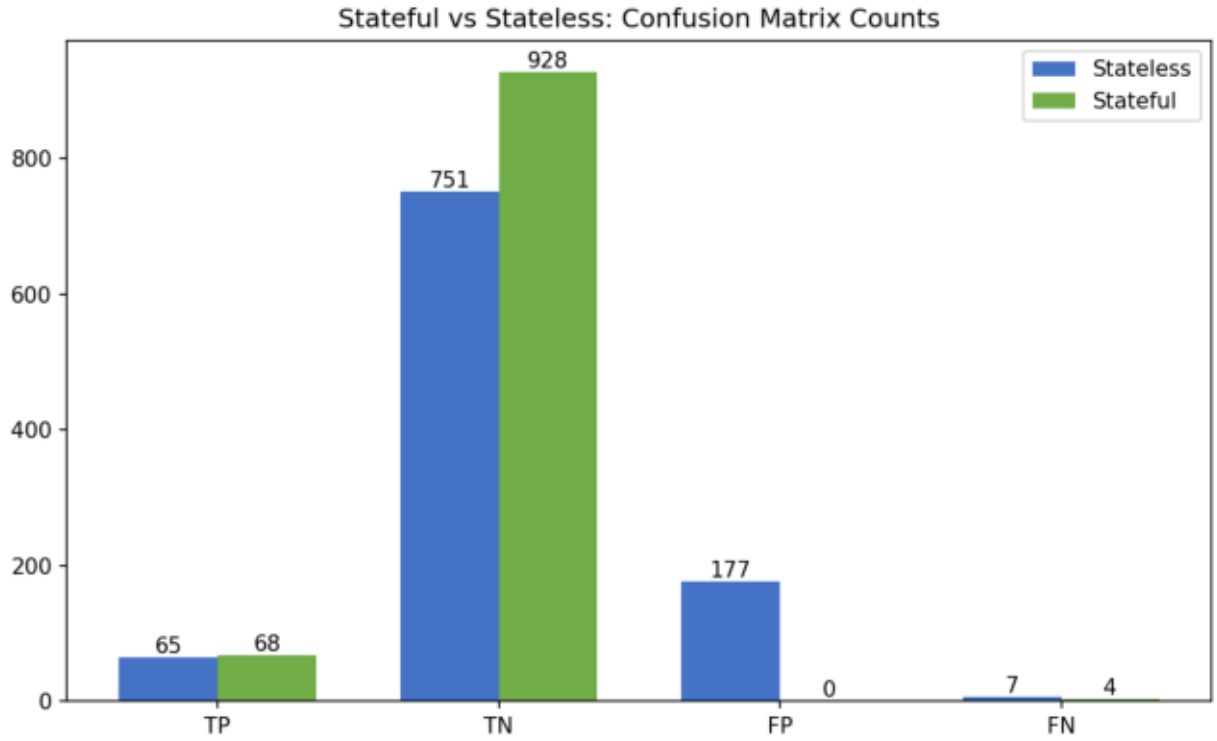
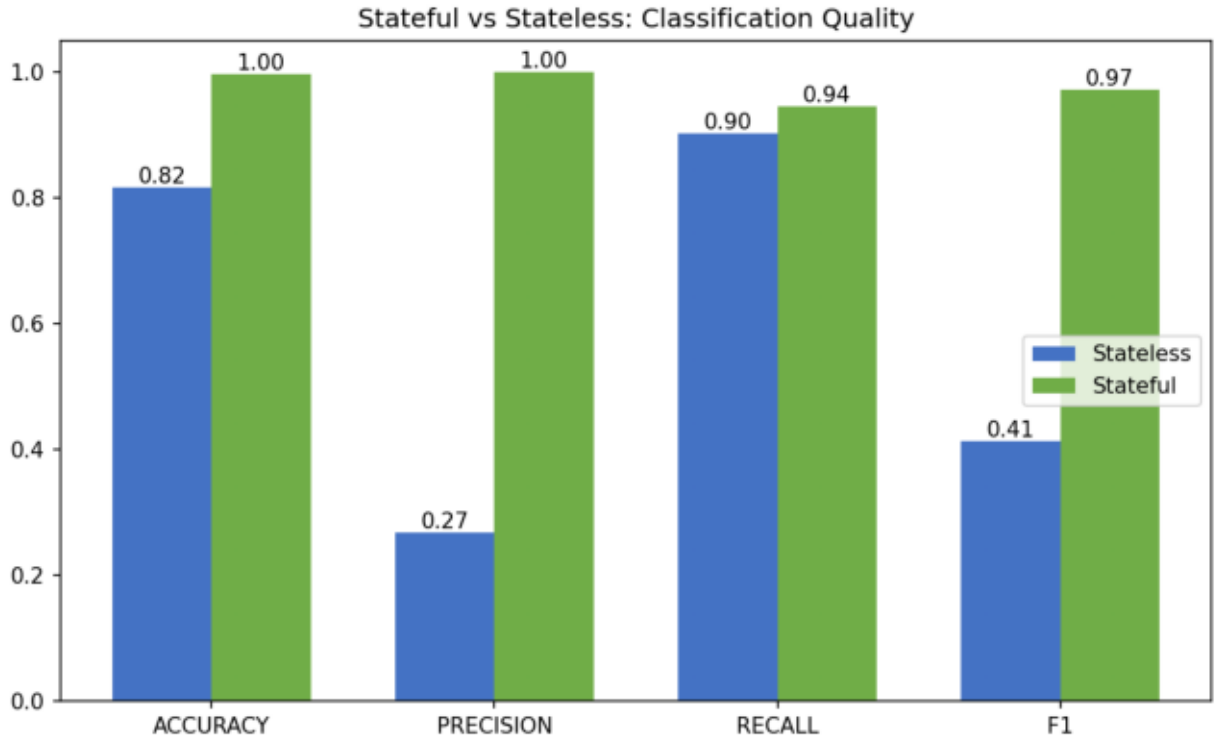
FIREWALL SIMULATION REPORT
packets: 1000
attack_ratio: 0.3
seed: 42
mode: comparison
live: False
delay: 0.2
demo: False
output_dir: None
compare: True
single_file: True
open_report: True

total: 1000
allowed: 932
blocked: 68
alerts: 28
tp: 68
tn: 928
fp: 0
fn: 4

Stateful reduced false positives from 177 to 0.


Stateful vs Stateless: Allowed, Blocked and Alerts
Stateless
Stateful
800
600
400
200
0
758
932
242
68
28
28
ALLOWED
BLOCKED
ALERTS


Stateful vs Stateless: Confusion Matrix Counts
Stateless
Stateful
800
600
400
200
0
65
68
751
928
177
0
7
4
TP
TN
FP
FN


Stateful vs Stateless: Classification Quality
1.0
0.8
0.6
0.4
0.2
0.0
0.82
1.00
0.27
1.00
0.90
0.94
0.41
0.97
Stateless
Stateful
ACCURACY
PRECISION
RECALL
F1

CONSOLE OUTPUT
Stateful vs Stateless comparison

Mode	Tp	Tn	Fp	Fn	Accuracy	Precision	Recall	F1
stateless	65	751	177	7	0.816	0.269	0.903	0.414
stateful	68	928	0	4	0.996	1.000	0.944	0.971